

Untitled

June 12, 2025

```
[ ]: import pandas as pd
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```
[ ]: df = pd.read_csv(r"C:\Users\HP\OneDrive\Desktop\supermarket_analysis\data\superstore.csv")
```

```
[ ]: df.head(3)
```

```
[ ]:
```

	Category	City	Country	Customer.ID	Customer.Name	\
0	Office Supplies	Los Angeles	United States	LS-172304	Lycoris Saunders	
1	Office Supplies	Los Angeles	United States	MV-174854	Mark Van Huff	
2	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	

	Discount	Market			Order.Date	Order.ID	...	Sales	\
0	0.0	US	1	2011-01-07	00:00:00.000	CA-2011-130813	...	19	
1	0.0	US	1	2011-01-21	00:00:00.000	CA-2011-148614	...	19	
2	0.0	US	1	2011-08-05	00:00:00.000	CA-2011-118962	...	21	

	Segment		Ship.Date	Ship.Mode	Shipping.Cost	\
0	Consumer	2011-01-09	00:00:00.000	Second Class	4.37	
1	Consumer	2011-01-26	00:00:00.000	Standard Class	0.94	
2	Consumer	2011-08-09	00:00:00.000	Standard Class	1.81	

	State	Sub.Category	Year	Market2	weeknum
0	California	Paper	2011	North America	2
1	California	Paper	2011	North America	4
2	California	Paper	2011	North America	32

[3 rows x 27 columns]

```
[ ]: df.shape
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 27 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Category              51290 non-null  object
```

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1  City                51290 non-null object
2  Country             51290 non-null object
3  Customer.ID         51290 non-null object
4  Customer.Name       51290 non-null object
5  Discount            51290 non-null float64
6  Market              51290 non-null object
7                      51290 non-null int64
8  Order.Date          51290 non-null object
9  Order.ID            51290 non-null object
10 Order.Priority      51290 non-null object
11 Product.ID          51290 non-null object
12 Product.Name        51290 non-null object
13 Profit              51290 non-null float64
14 Quantity            51290 non-null int64
15 Region              51290 non-null object
16 Row.ID              51290 non-null int64
17 Sales                51290 non-null int64
18 Segment             51290 non-null object
19 Ship.Date           51290 non-null object
20 Ship.Mode            51290 non-null object
21 Shipping.Cost        51290 non-null float64
22 State                51290 non-null object
23 Sub.Category         51290 non-null object
24 Year                 51290 non-null int64
25 Market2             51290 non-null object
26 weeknum              51290 non-null int64
dtypes: float64(3), int64(6), object(18)
memory usage: 10.6+ MB

```

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[ ]: drops=['Row.ID', ' ', 'Market2', 'weeknum']
df.drop(columns=drops,inplace=True,errors='ignore')
```

```
[ ]: df.shape
```

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[ ]: (51290, 23)
```

```
[ ]: df.columns=(df.columns.str.strip().str.lower().str.replace(".", "_").str.
↳replace(" ", "_"))
```

```
[ ]: df['order_date']=pd.to_datetime(df['order_date'])
df['ship_date']=pd.to_datetime(df['ship_date'])
```

```
[ ]: df.to_csv("../data/superstore_clean.csv",index=False)
print("Cleaned csv saved!!")
```

Cleaned csv saved!!

```
[ ]: !pip install psycopg2-binary sqlalchemy
```

Requirement already satisfied: psycopg2-binary in
c:\users\hp\anaconda3\lib\site-packages (2.9.10)
Requirement already satisfied: sqlalchemy in c:\users\hp\anaconda3\lib\site-
packages (1.4.54)
Requirement already satisfied: greenlet!=0.4.17 in
c:\users\hp\anaconda3\lib\site-packages (from sqlalchemy) (3.0.1)

```
[ ]: import psycopg2

conn = psycopg2.connect(
    dbname="superstore_db",
    user="postgres",
    password="learnsql321@",
    host="localhost",
    port="5432"
)
cursor=conn.cursor()
print(" psycopg2 connection successful!")
```

psycopg2 connection successful!

```
[ ]: create_table_sql = """
CREATE TABLE IF NOT EXISTS superstore (
    id SERIAL PRIMARY KEY,
    category TEXT,
    city TEXT,
    country TEXT,
    customer_id TEXT,
    customer_name TEXT,
    discount FLOAT,
    market TEXT,
    order_date DATE,
    order_id TEXT,
    order_priority TEXT,
    product_id TEXT,
    product_name TEXT,
    profit FLOAT,
    quantity INTEGER,
    region TEXT,
    sales FLOAT,
    segment TEXT,
    ship_date DATE,
    ship_mode TEXT,
    shipping_cost FLOAT,
    state TEXT,
    sub_category TEXT,
    year INTEGER
);
```

```

"""

cursor.execute(create_table_sql)
conn.commit()
print(" Table 'superstore' created successfully.")

```

Table 'superstore' created successfully.

```

[ ]: from tqdm import tqdm

columns = df.columns.tolist()

insert_query = f"""
INSERT INTO superstore ({", ".join(columns)})
VALUES ({", ".join(["%s" * len(columns)]})
"""

for row in tqdm(df.itertuples(index=False, name=None), total=len(df)):
    cursor.execute(insert_query, row)

conn.commit()
print(" All rows inserted into PostgreSQL successfully!")

```

```

100%|          |
51290/51290 [00:25<00:00, 2017.98it/s]

All rows inserted into PostgreSQL successfully!

```

```

[ ]: cursor.execute("Select * from superstore limit 5;")
rows=cursor.fetchall()
for row in rows:
    print(row, "\n")

```

```

(1, 'Office Supplies', 'Los Angeles', 'United States', 'LS-172304', 'Lycoris
Saunders', 0.0, 'US', datetime.date(2011, 1, 7), 'CA-2011-130813', 'High', 'OFF-
PA-10002005', 'Xerox 225', 9.3312, 3, 'West', 19.0, 'Consumer',
datetime.date(2011, 1, 9), 'Second Class', 4.37, 'California', 'Paper', 2011)

```

```

(2, 'Office Supplies', 'Los Angeles', 'United States', 'MV-174854', 'Mark Van
Huff', 0.0, 'US', datetime.date(2011, 1, 21), 'CA-2011-148614', 'Medium', 'OFF-
PA-10002893', 'Wirebound Service Call Books, 5 1/2" x 4"', 9.2928, 2, 'West',
19.0, 'Consumer', datetime.date(2011, 1, 26), 'Standard Class', 0.94,
'California', 'Paper', 2011)

```

```

(3, 'Office Supplies', 'Los Angeles', 'United States', 'CS-121304', 'Chad
Sievert', 0.0, 'US', datetime.date(2011, 8, 5), 'CA-2011-118962', 'Medium',
'OFF-PA-10000659', 'Adams Phone Message Book, Professional, 400 Message
Capacity, 5 3/6" x 11"', 9.8418, 3, 'West', 21.0, 'Consumer',

```

`datetime.date(2011, 8, 9), 'Standard Class', 1.81, 'California', 'Paper', 2011)`

`(4, 'Office Supplies', 'Los Angeles', 'United States', 'CS-121304', 'Chad Sievert', 0.0, 'US', datetime.date(2011, 8, 5), 'CA-2011-118962', 'Medium', 'OFF-PA-10001144', 'Xerox 1913', 53.2608, 2, 'West', 111.0, 'Consumer', datetime.date(2011, 8, 9), 'Standard Class', 4.59, 'California', 'Paper', 2011)`

`(5, 'Office Supplies', 'Los Angeles', 'United States', 'AP-109154', 'Arthur Prichep', 0.0, 'US', datetime.date(2011, 9, 29), 'CA-2011-146969', 'High', 'OFF-PA-10002105', 'Xerox 223', 3.1104, 1, 'West', 6.0, 'Consumer', datetime.date(2011, 10, 3), 'Standard Class', 1.32, 'California', 'Paper', 2011)`